

Case Study:

Preference-Based Housing Assignment in Uruguayan Cooperatives

Housing cooperatives in Uruguay play a crucial role in providing affordable housing to their members. Traditionally, once a new cooperative building was completed, the assignment of individual housing units was conducted through a random draw. This method, while simple, ignored individual preferences such as sunlight orientation, floor level, and proximity to shared facilities. Recognizing the potential for improved member satisfaction, researchers developed an alternative assignment method based on **mixed-integer programming (MIP)**.

The Challenge

In Uruguayan housing cooperatives, every member is entitled to a housing unit. Since the assignment process happens only once per building, it is crucial to ensure fairness while maximizing overall satisfaction. The existing lottery-based system failed to account for members' preferences, sometimes leading to dissatisfaction and informal exchanges of units after the assignment. The challenge was to develop a new assignment process that:

- Ensured fairness among all members,
- Incorporated individual preferences,
- Was easy to explain and implement,
- Could be executed in real time during cooperative meetings.

The Solution: A Two-Stage Optimization Model

A team of researchers proposed a **two-stage mixed-integer programming (MIP) model** to address these concerns. The assignment problem was formulated with the following structure:

Stage 1 – Ensuring Fairness: The model maximizes the satisfaction of the least satisfied member. This ensures that no one is assigned an extremely undesirable unit.

Stage 2 – Optimizing Global Satisfaction: Given the fairness constraints from Stage 1, the model then seeks to optimize the total satisfaction across all cooperative members.

The assignment process was implemented using a software tool called **MTAV (Mejor Tecnología de Asignación de Viviendas – A Better Technology for Housing Assignment)**, designed to be transparent and user-friendly.

Implementation and Results

The first real-world test of the model occurred in **2016** with the **Virazón housing cooperative**, consisting of **50 members**. Before deployment, the cooperative members reviewed and voted to approve the new method. Their individual housing preferences were collected and input into the system.

On the day of the assignment:

- The housing allocation was carried out in **real-time** during a cooperative meeting.

- The results were projected on a large screen for transparency.
- A notary was present to certify the fairness of the process.
- The optimization model took only a few **seconds** to compute the assignments.

The outcomes were striking:

- **Four-bedroom apartments:** Both members received their first-choice units.
- **Three-bedroom apartments:** More than 37% of members received their first-choice unit, and the average satisfaction ranking was significantly better than a random assignment.
- **Two-bedroom apartments:** Over **half** of the members received one of their top four choices, with all assignments being significantly better than expected under a random draw.

The cooperative members overwhelmingly supported the new system, reporting higher satisfaction compared to previous assignment methods.

Broader Impact

Since the successful implementation at Virazón, **over 30 housing cooperatives** in Uruguay have adopted the model. Further developments have included:

- The introduction of a **graphical user interface** to make data input more intuitive.
- Support for **partial order preferences**, allowing members to indicate that they have no strong preference between certain units.
- Exploration of additional fairness criteria and trade-offs between individual and collective satisfaction.

Beyond its technical achievements, this case study highlights how **operations research techniques** can be applied to solve real-world social challenges. By integrating mathematical optimization with participatory decision-making, the project has significantly improved the way housing is assigned in cooperative communities, setting a precedent for future applications in housing policy and beyond.

Optimization models

The optimization model required the following input parameters and variables:

Sets:

- I : Set of cooperative members needing housing assignments.
- J : Set of available housing units.

Parameters:

- p_{ij} : Preference score of member i for housing unit j , where lower values indicate higher preference (e.g., 1 = most preferred).
- Some versions of the model supported **partial order preferences**, allowing members to express equal preference for multiple units.

Decision Variables:

- $x_{ij} \in \{0, 1\}$: Binary variable indicating whether member i is assigned to unit j :

$$x_{ij} = \begin{cases} 1, & \text{if member } i \text{ is assigned to unit } j \\ 0, & \text{otherwise} \end{cases}$$

Two-Stage Optimization Approach

Stage 1: Ensuring Fairness

- Define S as the satisfaction level of the least satisfied member:

$$S = \min_{i \in I} \sum_{j \in J} p_{ij} x_{ij}$$

- Ensure each member is assigned exactly one unit and each unit is assigned to exactly one member.

This is achieved by the model

$$S = \min z \quad (5.9)$$

$$\text{s.t. } z \geq \sum_{j \in J} p_{ij} x_{ij}, \quad \forall i \in I \quad (5.10)$$

$$\sum_{j \in J} x_{ij} = 1, \quad \forall i \in I \quad (5.11)$$

$$\sum_{i \in I} x_{ij} = 1, \quad \forall j \in J \quad (5.12)$$

$$x_{ij} \in \{0, 1\}, \quad \forall i \in I, j \in J \quad (5.13)$$

$$z \in \mathbb{R} \quad (5.14)$$

Stage 2: Optimizing Global Satisfaction

- Given the fairness threshold S from Stage 1, optimize total satisfaction:

$$\min \sum_{i \in I} \sum_{j \in J} p_{ij} x_{ij}$$

- Subject to the fairness constraint:

$$\sum_{j \in J} p_{ij} x_{ij} \leq S, \quad \forall i \in I$$

- Ensures that no member receives a unit less desirable than the worst outcome allowed by Stage 1.

This is achieved by the model

$$\min \sum_{i \in I} \sum_{j \in J} p_{ij} x_{ij} \quad (5.15)$$

$$\text{s.t. } \sum_{j \in J} x_{ij} = 1, \quad \forall i \in I \quad (5.16)$$

$$\sum_{i \in I} x_{ij} = 1, \quad \forall j \in J \quad (5.17)$$

$$\sum_{j \in J} p_{ij} x_{ij} \leq S, \quad \forall i \in I \quad (5.18)$$

$$x_{ij} \in \{0, 1\}, \quad \forall i \in I, j \in J \quad (5.19)$$

Implementation Notes

- The model was formulated as a **mixed-integer programming (MIP)** model.
- The assignment was computed in **real-time** during cooperative meetings.
- The **MTAV** software was used to collect user preferences and compute assignments efficiently.
- The system ensured transparency, allowing cooperative members to verify their preferences before the final optimization.

References

- [IFORS Newsletter December 2024 - Section 4: OR Impact](#)
- M. Prino, E. Sánchez, H. Cancela. “Optimal distribution of habitational units in a cooperative: A mathematical application to optimize satisfaction” (text in Spanish). Proceedings of CLEI 2016 (Conferencia Latinoamericana de Informática), 2016. <https://doi.org/10.1109/CLEI.2016.7833357>

5.4 Modeling Tricks

5.4.1. Maximizing a minimum

When the constraints could be general, we will write $x \in X$ to define general constraints. For instance, we could have $X = \{x \in \mathbb{R}^n : Ax \leq b\}$ or $X = \{x \in \mathbb{R}^n : Ax \leq b, x \in \mathbb{Z}^n\}$ or many other possibilities.

Consider the problem

$$\begin{aligned} \max \quad & \min\{x_1, \dots, x_n\} \\ \text{such that} \quad & x \in X \end{aligned}$$

Having the minimum on the inside is inconvenient. To remove this, we just define a new variable y and enforce that $y \leq x_i$ and then we maximize y . Since we are maximizing y , it will take the value of the smallest x_i . Thus, we can recast the problem as

$$\begin{aligned} \max \quad & y \\ \text{such that} \quad & y \leq x_i \text{ for } i = 1, \dots, n \\ & x \in X \end{aligned}$$